

HONG KONG BAPTIST UNIVERSITY

COURSE OUTLINE

1. COURSE TITLE

Blockchain and Cryptocurrencies

2. COURSE CODE

COMP7200

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

Nil

6. MEDIUM OF INSTRUCTION

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7. AIMS & OBJECTIVES

To introduce the concepts and applications of the blockchain technology; To introduce concepts and applications of two cryptocurrencies, Bitcoin and Ethereum; To explain the latest techniques of cryptocurrency and permissionless as well as permissioned blockchains; To conduct case studies to apply the blockchain technology to real-world applications.

8. COURSE CONTENT

I. Introduction to Blockchains and Cryptography

- Overview of Blockchain
- Types of Blockchains
- Blockchain Applications
- Cryptographic Primitives
- Distributed Systems
- Consensus Protocols

II. Cryptocurrency

- Overview of Cryptocurrency
- Bitcoin basics
- Bitcoin implementation details
- Bitcoin's variants

III. Cryptocurrency on Permissionless Blockchains

- Ethereum basics
- Ethereum implementation details

IV. Smart Contract and Decentralized Applications

- Ethereum Virtual Machine
- Smart Contract, Dapp, DAO, Token, etc
- Developing Decentralized Applications in Ethereum

V. Permissioned Blockchain

- Hyperledger Fabric
- Applications of Hyperledger Fabric

VI. Advanced Topics on Blockchain and Cryptocurrencies

- Decentralized Finance (DeFi)
- Anonymity
- Privacy Protection
- Scalability: Lightning Network, Sharding, etc

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Compare different types of blockchains and cryptocurrencies
CILO 2	Identify the concepts and techniques of the cryptocurrency and consensus protocols
CILO 3	Describe the concepts and techniques of the smart contract and hyperledger
CILO 4	Investigate emerging techniques of the advanced topics on blockchain and cryptocurrencies
CILO 5	Design appropriate solutions to blockchain applications
CILO 6	Develop real-world applications using the blockchain technology

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1-4	Students will acquire the knowledge and skills on the blockchain technology and cryptocurrencies through lectures.
2-5	Students will learn the development skills of blockchain applications

	especially in fungible tokens and non-fungible tokens (NFT) via tutorials, assignments, quizzes, and guided labs.
5-6	Students will work on a group project to gain application development experience.

11. **ASSESSMENT METHODS (AMs)**

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Continuous Assessment	40 %	1-6	Assignments, quizzes and laboratories are designed to evaluate students' mastery of cryptocurrency concepts and techniques. A group project is used to assess their problem solving and application development skills.
Examination	60 %	1-5	Final examination questions are designed to assess how well students understand and utilize the knowledge acquired.

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